

Science, Physics, and Mathematics as Boundary Discipline

How to use precision without turning it into decoration

Science, Physics, and Mathematics as Boundary Discipline

How to use precision without turning it into decoration

Science, physics, and mathematics matter to the theory not because they grant a halo, but because they can say “this works here” and “this remains unknown there.” Boundary discipline is the ability to use a precise concept without loading it with an untested claim. It is the opposite of formula mysticism.

The common failure is replacing precision with legitimacy. Once a formula appears, a reader may think the philosophical claim has been proven. That is dangerous. A formula in this theory is a relation map, not a prediction engine. Without units, measurement, domain of validity, and possible falsification, it must not be treated as a scientific formula.

Boundary discipline = definition × domain of validity × distinction between metaphor and model × source × possibility of critique / borrowed authority × field-mixing × grand words × pretended proof. Every chapter mentioning QFT, Higgs, Bell, Kochen-Specker, Bohmian mechanics, contextuality, locality, or non-locality must pass this map.

Mathematics teaches the theory a special humility. It distinguishes possibility, proof, verification, computation, existence, construction, and decision. The theory is not proven by P vs NP, Gödel, or Turing. It uses them as language about limits of closure, verification, and systems trying to contain themselves.

Physics teaches another humility: a theory can be extremely precise within a domain and still partial. A model can work, fail at an edge, be extended, or become a limiting case of a deeper language. That is not a flaw; it is the way of science. The theory learns that a real optimum knows its domain of validity.

Boundary discipline also protects the living source. When scientific language is used as decoration, it exploits the work of scientists as an effect. When used carefully, it honors the distance they carried to turn not-knowing into partial responsible knowledge.

Source discipline

Working sources and caution: this chapter relies on the theory itself and on background sources checked outside the prompt: UNESCO on artificial intelligence in education, WHO on ethics and governance of AI for health, the NIST AI RMF for AI risk management, CERN/ATLAS on the Standard Model and the Higgs, NASA and

ESA/Planck on cosmology, the Stanford Encyclopedia of Philosophy on Bell, Kochen-Specker and contextuality, and Britannica/ISFDB for the literary identification of Harlan Ellison and *I Have No Mouth, and I Must Scream*. These sources are not treated as proofs of the theory; they are safeguards against empty borrowing of terms.